Presented by :
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Procurement Cost Optimization

Analyzing Spend, Cost Savings, and Supplier Performance

Project Overview

Business Context

The company manages procurement across multiple suppliers, with over 700 purchase orders between 2022 and 2023. Rising costs, supplier delays, and inconsistent quality have created challenges, while limited visibility into cost drivers and supplier performance hinders effective decision-making.

Business Objectives

Objective 1

Identify key cost drivers and spending patterns

Objective 2

Evaluate the effectiveness of price negotiations

Objective 3

Assess supplier performance (quality and delivery)

Data Overview

Dataset Summary

- 777 purchase orders (2022–2023)
- 5 suppliers across multiple procurement categories
- Categories include Raw Materials, MRO, Electronics, Packaging, and Office Supplies
- Each record represents a single procurement transaction

Key Variables & Metrics

- Pricing: Unit Price vs. Negotiated Price (cost savings analysis)
- Quantity: Total procurement volume
- Lead Time: Order date to delivery date (lead time analysis)
- Quality: Defective units (defect rate evaluation)
- Compliance: Supplier compliance status (Yes/No)

Analytical Framework

Spend Analysis

- Spend by supplier
- Spend by category

Cost Savings Analysis

- Unit vs negotiated price
- Cost reduction

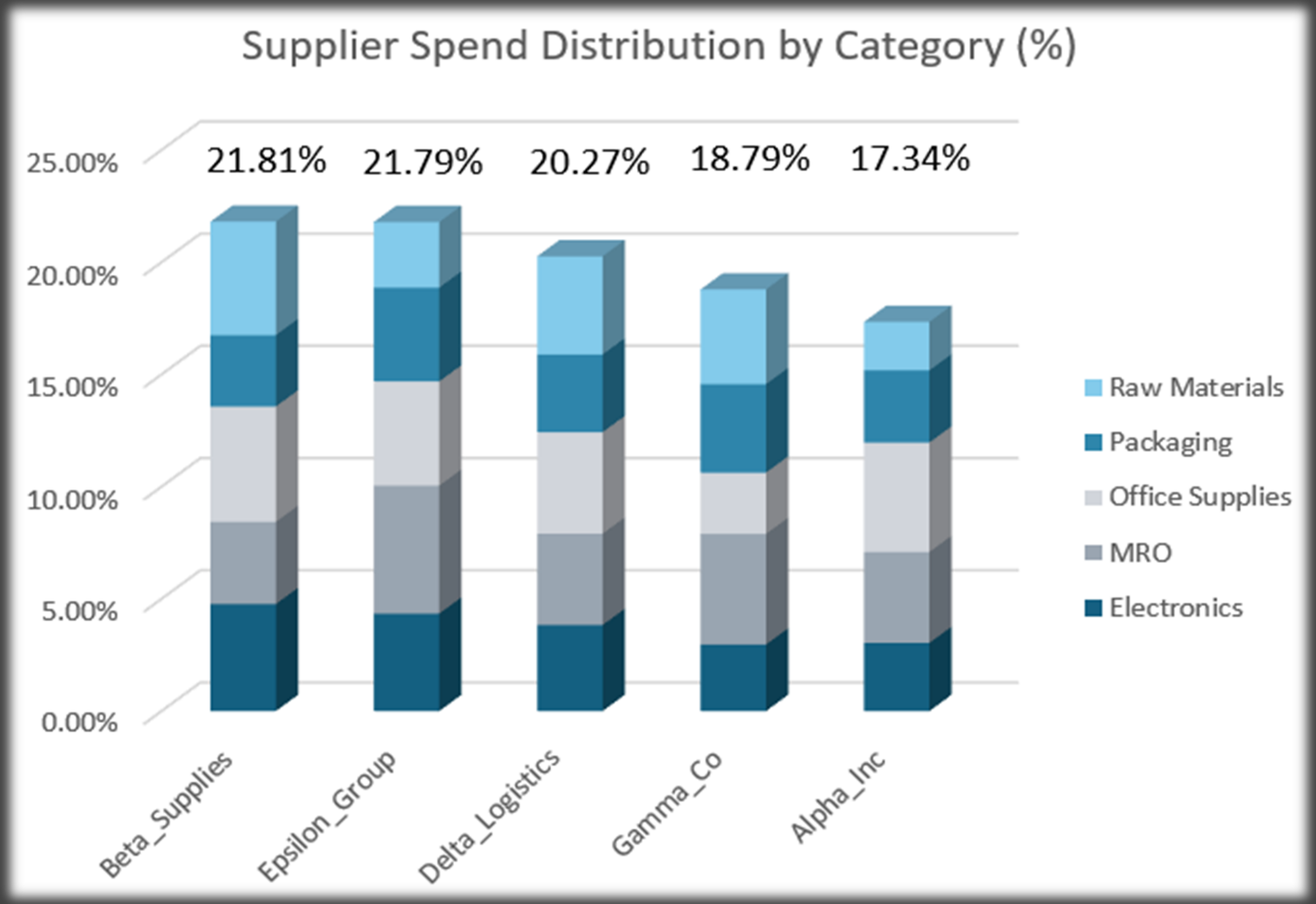
Supplier Performance Analysis

- Lead time
- Defect rate
- Compliance rate

Spend Analysis

Supplier Spend Distribution

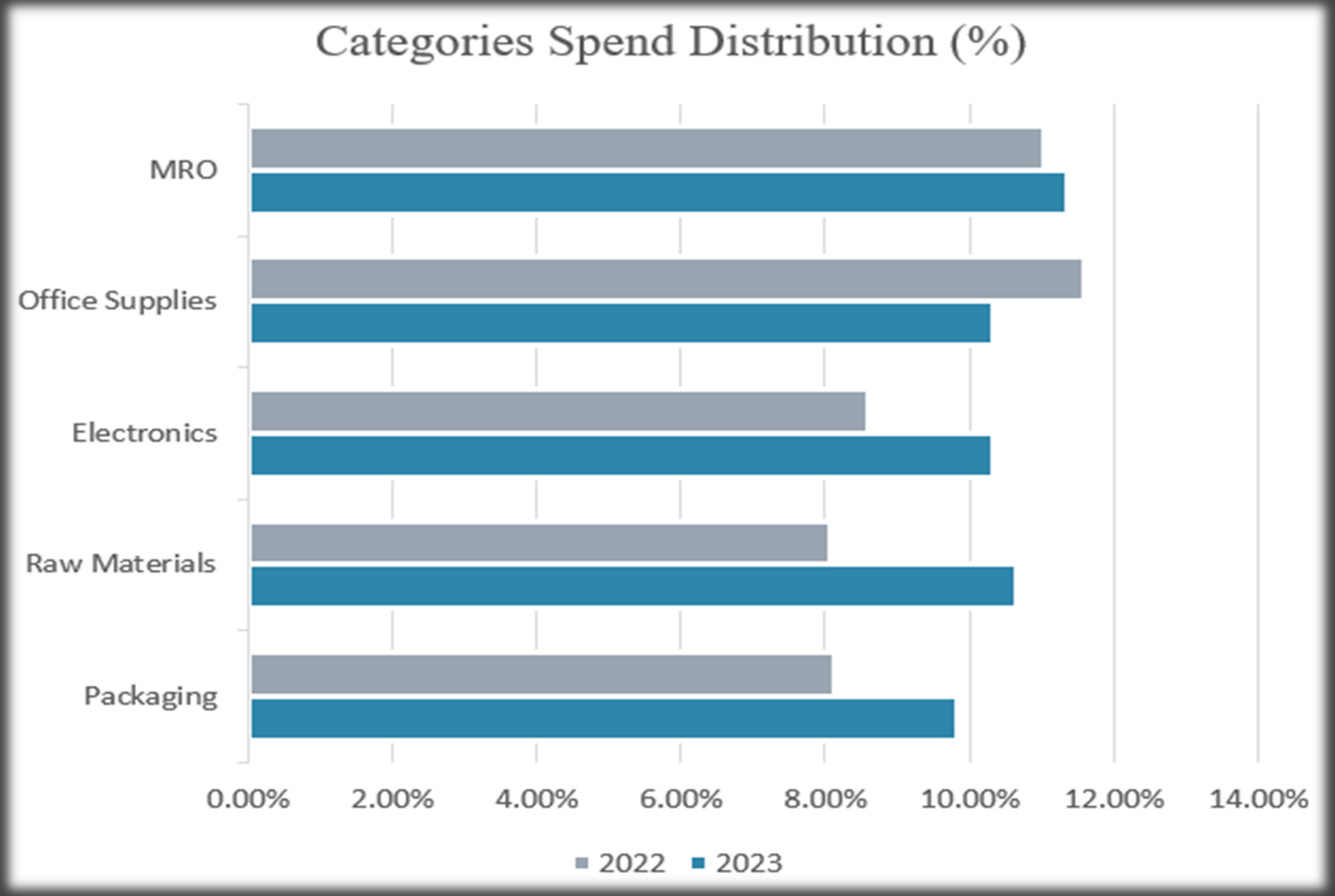
Supplier spend is relatively evenly distributed across vendors, indicating a balanced supplier structure.
This reduces concentration risk but may limit negotiation power and increase operational complexity.



Spend Analysis

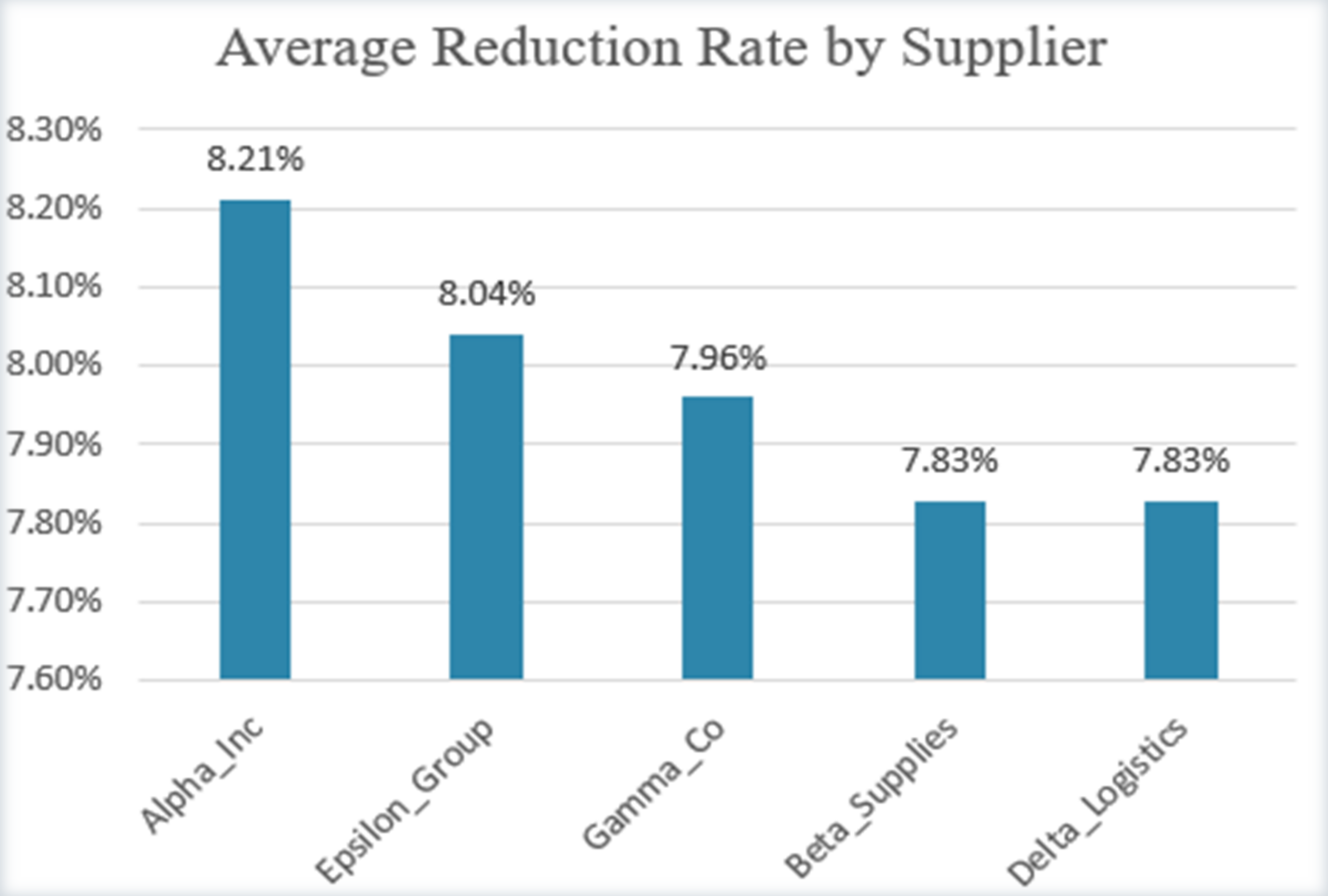
Category Spend Distribution

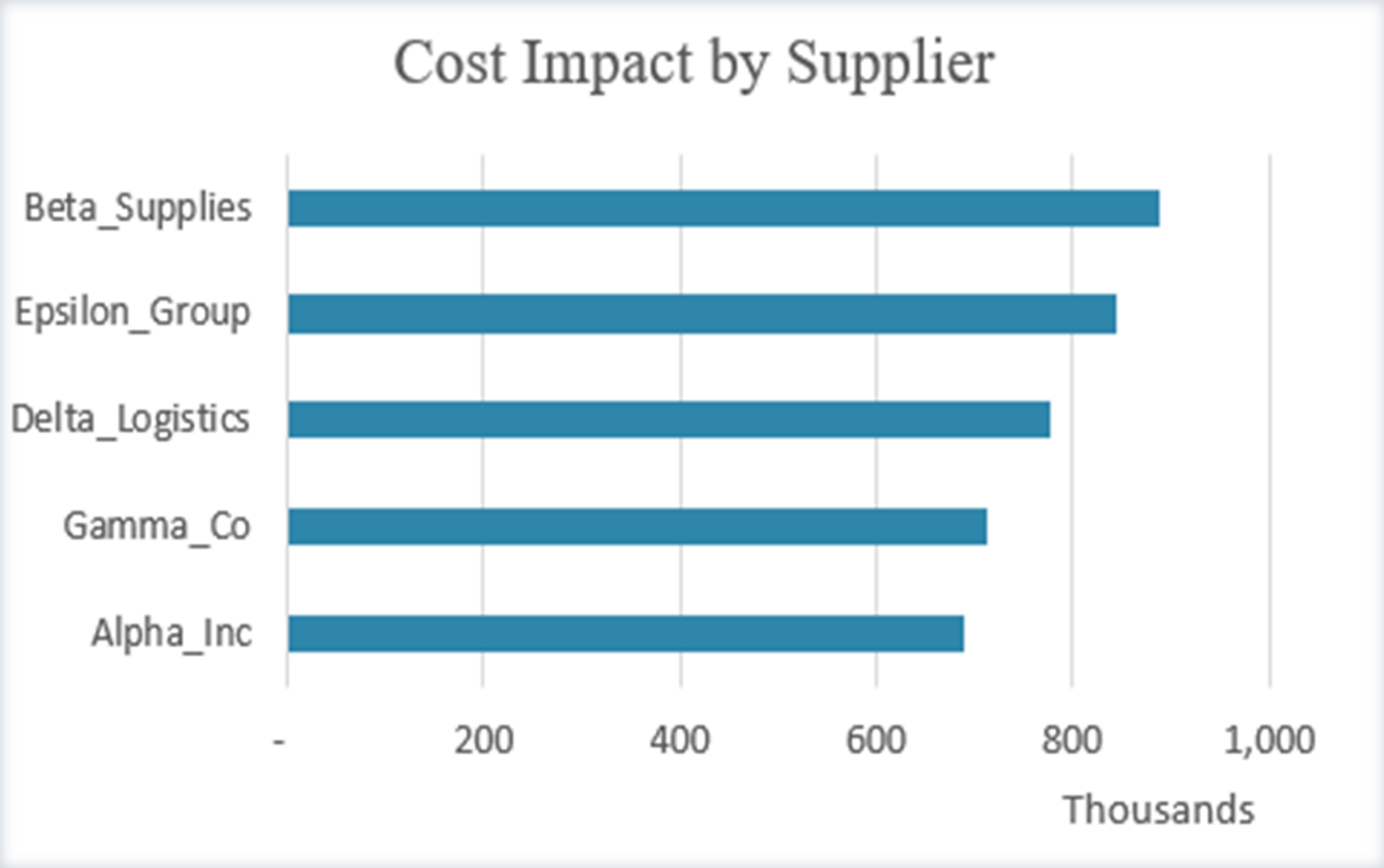
Spending increased in Raw Materials, Packaging, and Electronics, indicating higher production activity.
In contrast, Office Supplies declined, suggesting tighter control over indirect costs.



Cost Analysis

Supplier Cost Performance



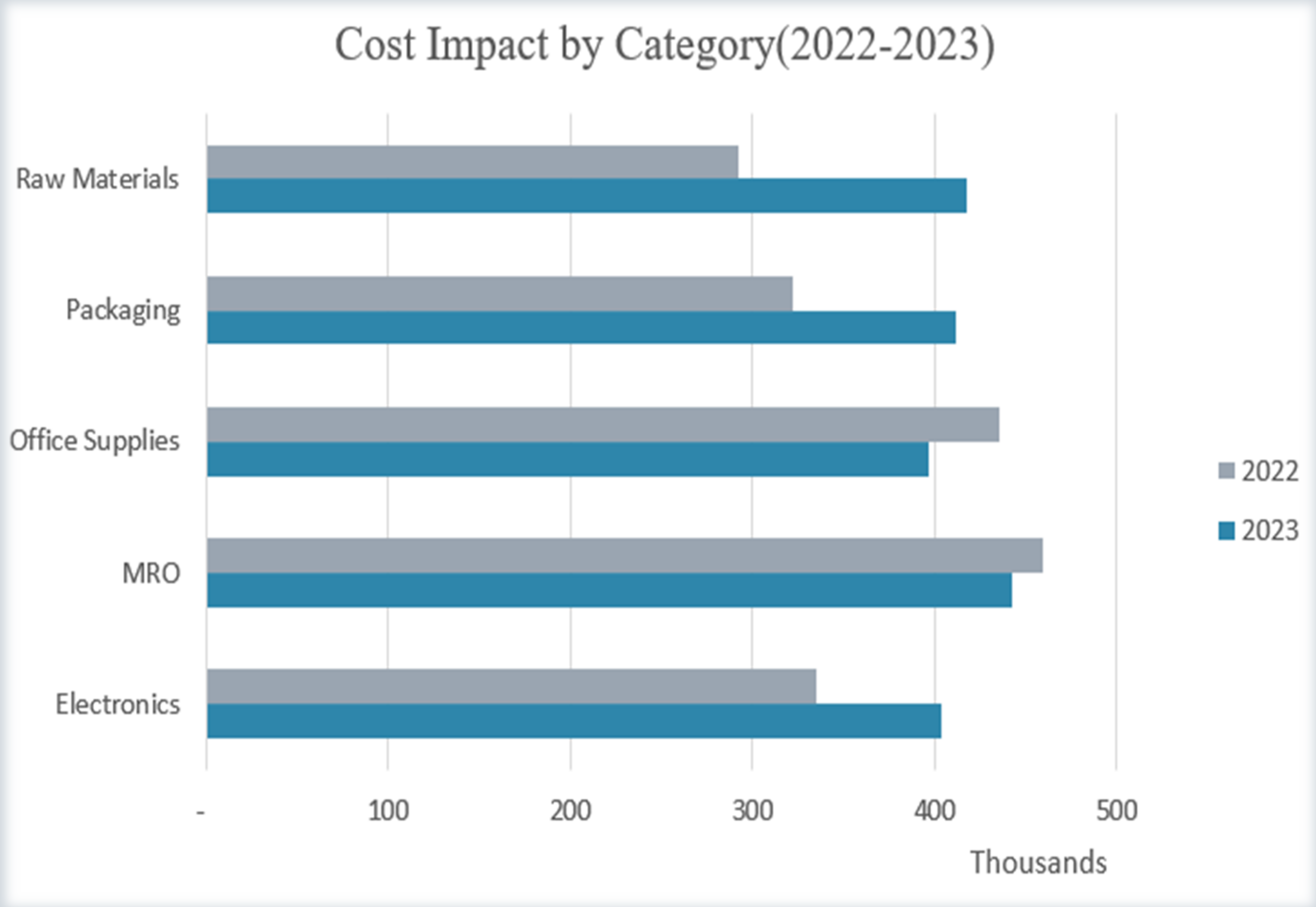


Alpha Inc shows strong negotiation but low impact due to lower spend.
In contrast, Beta and Epsilon drive the highest costs with only average discounts, indicating opportunities to improve negotiation on high-spend suppliers.

Cost Analysis

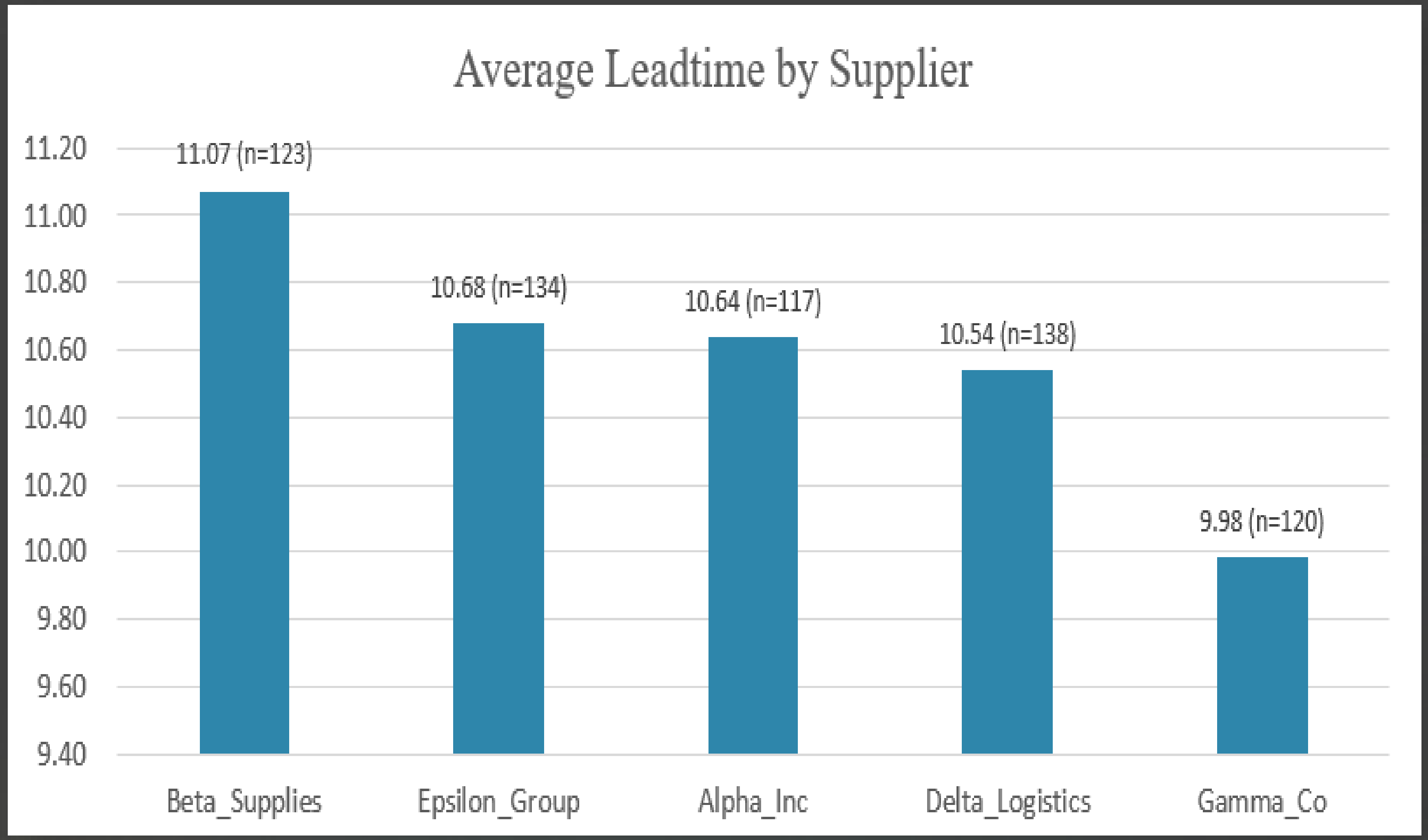
Cost Savings by Category

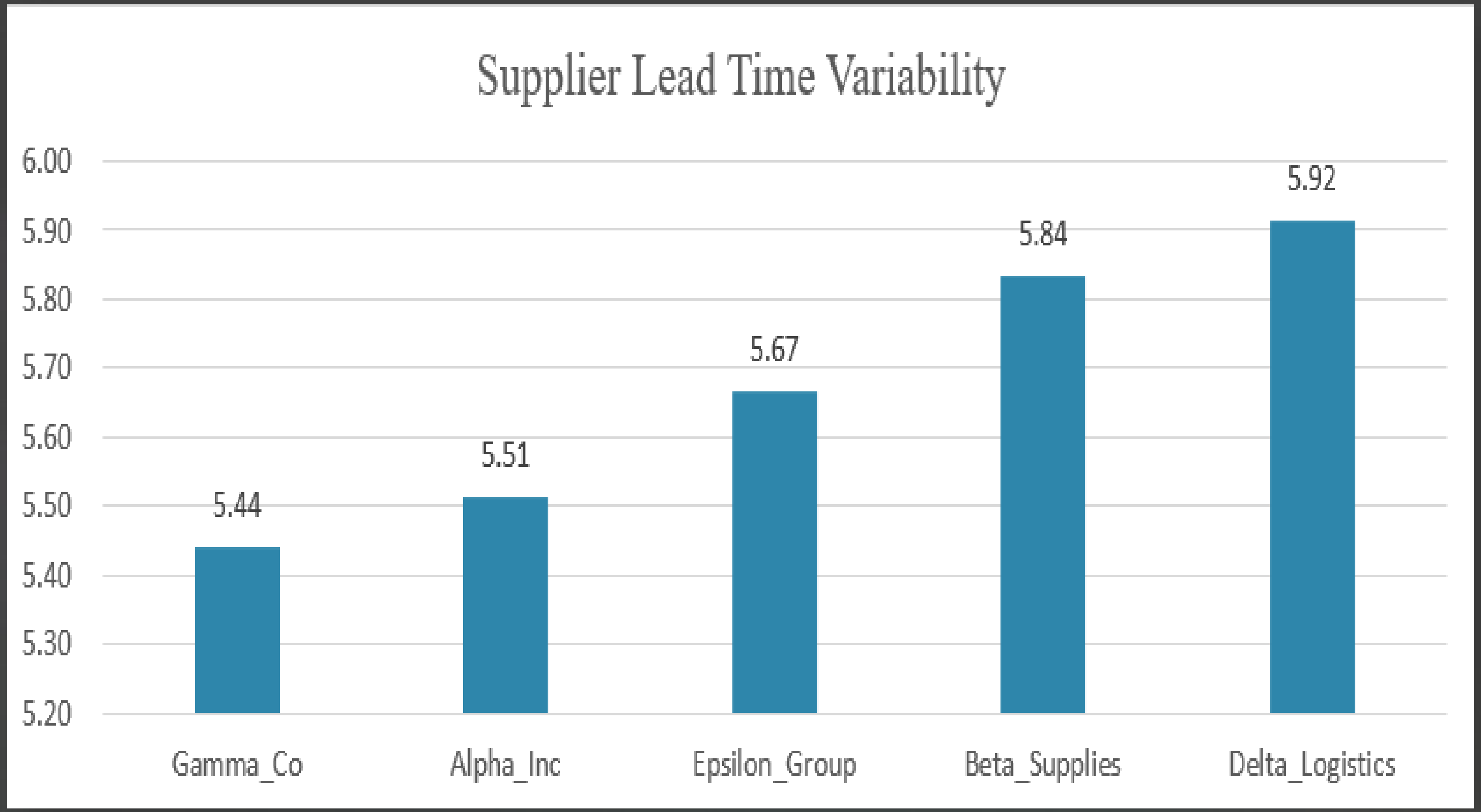
Cost savings increased (~12%), driven by Raw Materials, Packaging, and Electronics. These production-related categories contributed most to total savings, while MRO and Office Supplies had a limited impact.



Supplier Performance Analysis

Lead Time and Variability

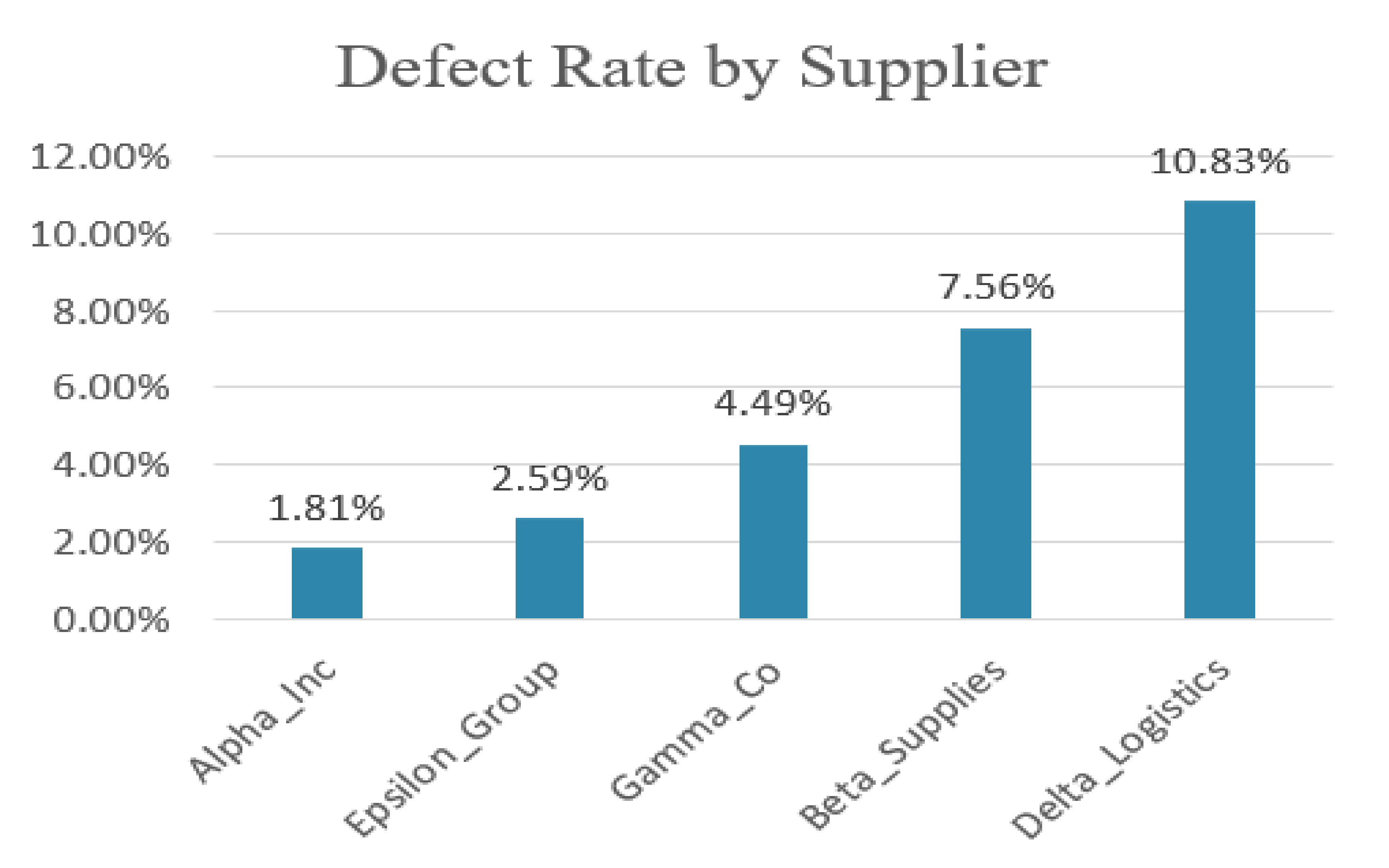


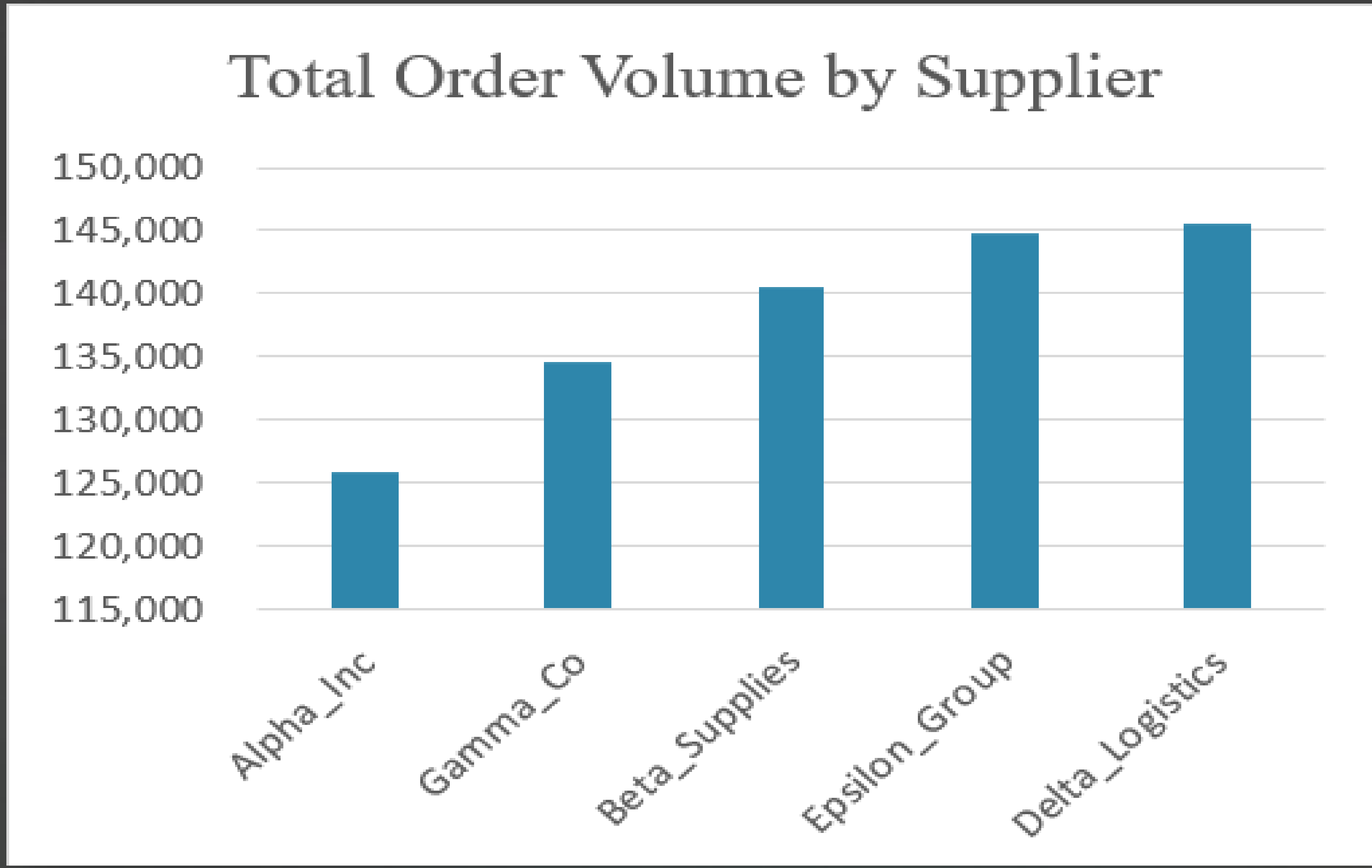


Gamma_Co leads in both lead time and consistency, indicating the strongest supplier performance. However, variability across suppliers remains relatively high, suggesting potential reliability risks.

Supplier Performance Analysis

Defect Rate



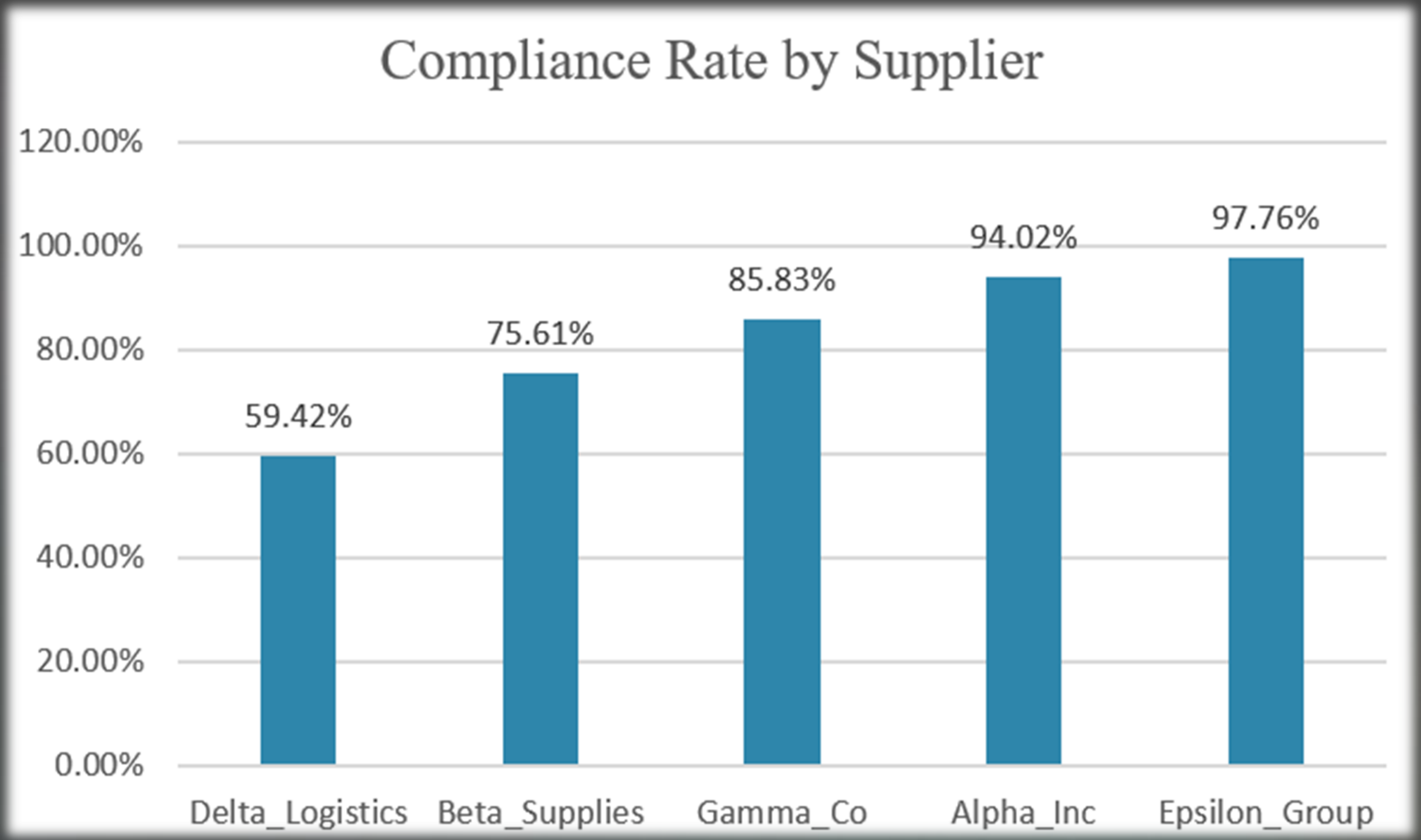


Delta Logistics combines the highest defect rate with high order volume, indicating a critical quality issue. In contrast, Epsilon Group achieves low defect rates at comparable volume, serving as a strong benchmark.

Supplier Performance Analysis

Compliance Rate

Delta Logistics shows the lowest compliance, indicating a potential issue that requires further investigation. In contrast, Epsilon Group achieves the highest compliance and serves as a strong benchmark.



Recommendations

COST FOCUS

Prioritize production-related categories (Raw Materials, Packaging, Electronics) for cost optimization through strategic sourcing and negotiation.

SUPPLIER STRATEGY

Reallocate volume from high-risk suppliers (Delta, Beta) to high-performing suppliers (Alpha, Epsilon), while strengthening negotiation on high-volume suppliers.

OPERATIONAL ACTIONS

- Optimize lead time for Beta and Epsilon
- Improve quality and compliance for Delta and Beta
- Rebalance allocation toward high-performing suppliers (Alpha, Gamma)

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Thank You

for your time and attention